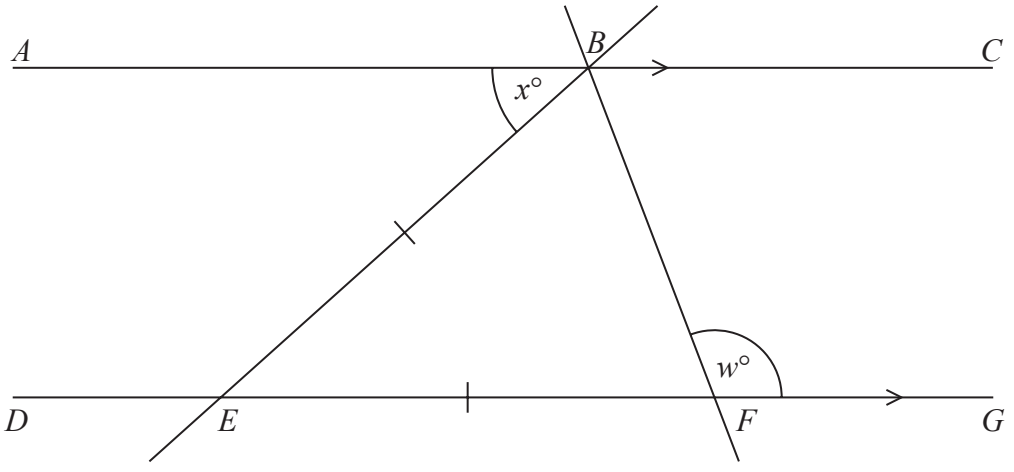
9



In the diagram ABC and $DEFG$ are parallel lines.
Angle $ABE = x°$
$EB = EF$

Show that $w = 90 + \frac{1}{2}x$

Give a reason for each stage of your working.

Angle BEF = x (alternate angles on parallel lines).

1 mark
Parallel-line angle

EB = EF, so triangle EBF is isosceles.

Angles EBF and EFB are equal base angles.

1 mark
Equal base angles

Each = (180° − x) ÷ 2 = 90° − ½x (triangle sum).

w + angle EFB = 180° (straight line).

1 mark
Complete w method

w = 180° − (90° − ½x) = 90° + ½x.

1 mark
All linked reasons

(Total for Question 9 is 4 marks)